

Practical# 04 Jenkins Pipeline to Pull Docker Image & Run Container

Objective

To create a Jenkins pipeline that pulls a Docker image from Docker Hub and runs it as a container.

Prerequisites

1. Jenkins installed and running 2. Docker Desktop installed and running 3. Git installed 4. Basic knowledge of Jenkins pipeline

Step 1: Verify Docker Installation

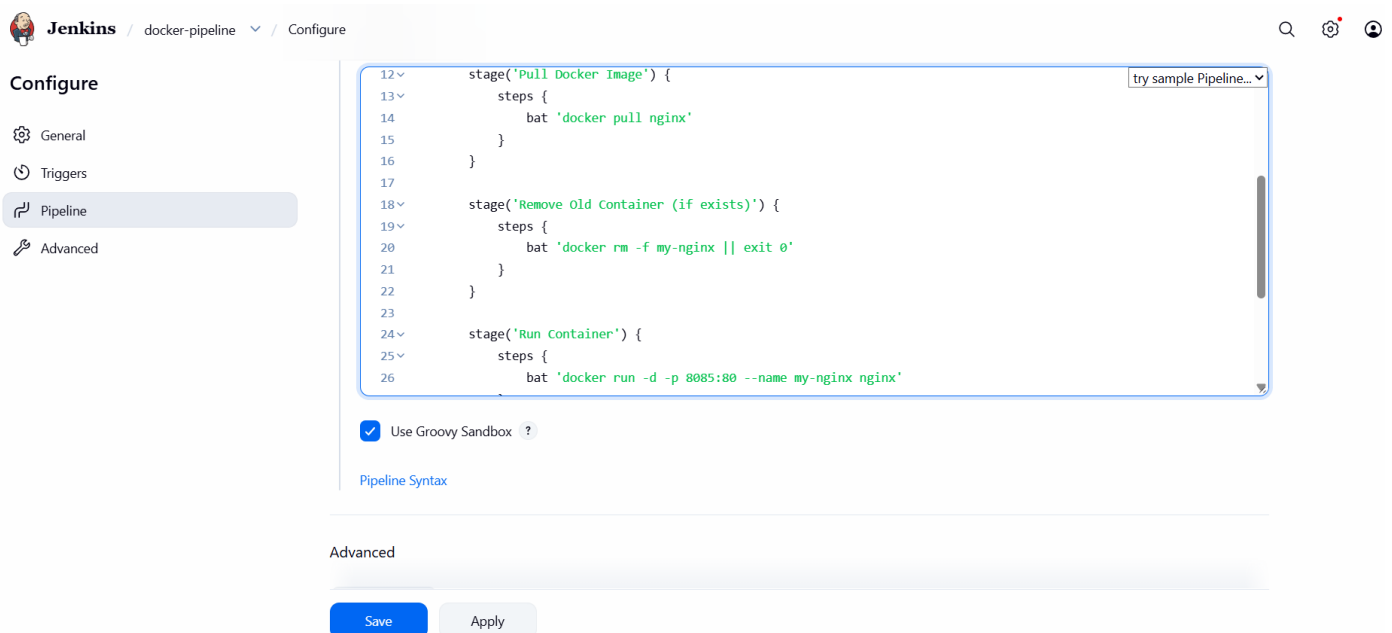
Run the following command in Jenkins or system terminal:

```
docker --version docker ps
```

Step 2: Create Jenkins Pipeline Job

1. Open Jenkins Dashboard 2. Click on 'New Item' 3. Enter name: docker-pipeline 4. Select Pipeline 5.

Click OK



The screenshot shows the Jenkins configuration page for a pipeline named 'docker-pipeline'. The left sidebar has tabs for 'General', 'Triggers', 'Pipeline', and 'Advanced', with 'Pipeline' selected. The main area displays a Groovy script for the pipeline. The script consists of three stages: 'Pull Docker Image', 'Remove Old Container (if exists)', and 'Run Container'. The 'Pull Docker Image' stage has a step to pull the 'nginx' image. The 'Remove Old Container (if exists)' stage has a step to remove the container if it exists. The 'Run Container' stage has a step to run the container with port 8085 mapped to 80 and named 'my-nginx'.

```
12 stage('Pull Docker Image') {
13   steps {
14     bat 'docker pull nginx'
15   }
16 }
17
18 stage('Remove Old Container (if exists)') {
19   steps {
20     bat 'docker rm -f my-nginx || exit 0'
21   }
22 }
23
24 stage('Run Container') {
25   steps {
26     bat 'docker run -d -p 8085:80 --name my-nginx nginx'
```

Below the script, there is a checkbox for 'Use Groovy Sandbox' which is checked. At the bottom, there are 'Save' and 'Apply' buttons.

Step 4: Add Pipeline Script

Write the content in the Pipeline Script inside it and then save it

Jenkins

docker-pipeline

#1

Status

Changes

Console Output

Delete build '#1'

Timings

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

✓ #1 (May 3, 2026, 6:51:35 PM)

Started by user Kavya Gupta

This run spent:

- 21 ms waiting;
- 14 sec build duration;
- 14 sec total from scheduled to completion.

</> No changes.

Add description

Keep this build forever

Started 29 sec ago

Took 14 sec

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Step 5: Run the Pipeline

Click on 'Build Now' and observe the pipeline stages.

Jenkins

docker-pipeline

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Stages

Pipeline Syntax

docker-pipeline

Permalinks

- Last build (#1), 1 min 4 sec ago
- Last stable build (#1), 1 min 4 sec ago
- Last successful build (#1), 1 min 4 sec ago
- Last completed build (#1), 1 min 4 sec ago

Add description

Builds >

Filter

Today

✓ #1 6:51 PM

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Step 6: Verify Output

Open browser and go to: <http://localhost:8085>

